Lesson notes · for you

Summaries that tell the truth

Data engineering · Level 2 · Lesson 4

Time
15 min

Materials
Your phone with Darsi, for the Python steps; Squared paper and a ruler

Learning goals

- Work out the mean, the median and the range.
- See how one extreme value drags the mean.
- Make an honest chart.

Activity: Line up, find the middle

1. Collect (3 min): Nine volunteers each write a number on a card, such as minutes to get to school.
2. Line up (4 min): The volunteers line up in order of their numbers. The person in the middle holds the median.
3. Add (4 min): Add all the numbers and divide by nine. Is the mean close to the median?
4. Change (4 min): Change one card to a huge number. Work out the mean and the median again: which one moved?

Use numbers nobody minds sharing, or let volunteers make them up.

Discussion

- Why didn't the median move when one number became huge?
- When would you report the median instead of the mean?
- What makes a chart honest?

Check yourself

- Write five numbers whose mean and median are far apart, and say which one you'd report.

Try it this week
Find an average in an advert or the news. Ask: is it a mean or a median? And what could be dragging it?

Worksheet

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Name _____ Class _____ Date _____

1 **Which of these makes a bar chart unfair?**

◯ A label under each bar ◯ A title at the top ◯ Bars that don't start at zero

2 **Complete the sentences.**

The _____ adds the numbers and divides by how many. The _____ is the middle one once they're sorted. The _____ is the biggest minus the smallest.

Word bank: total · mean · median · range · mode

3 **Five friends' data use, in GB. What does this program print?**

```
data = [1, 1, 2, 3, 9]
print(sum(data) / len(data))
print(sorted(data)[2])
```

◯ 3.2, then 2 ◯ 2, then 3.2 ◯ 3.2, then 3

4 **One mark was typed as 150 instead of 15. The program prints the mean, then tries to print the median, but it takes the middle of the unsorted list. Fix it so it prints 40.0, then 13.**

```
marks = [12, 150, 11, 14, 13]
print(sum(marks) / len(marks))
print(marks[2])
```

Write your program:

5 **When is the median a better summary than the mean? Choose all that apply.**

☐ House prices in a town with one mansion ☐ Data use, where a few people use a lot

☐ Five classmates' heights, all close together ☐ Marks with one typed as 150

✎ **Think of an average you've heard: in an advert, the news or at school. Was it a mean or a median? What might be dragging it?**

Check your answers

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1 **Bars that don't start at zero**

Cutting the axis makes small differences look huge.

2 **The mean adds the numbers and divides by how many. The median is the middle one once they're sorted. The range is the biggest minus the smallest.**

Mean, median and range sum up the same numbers in three different ways: the typical share, the middle value, and how spread out they are.

3 **3.2, then 2**

sum(data) / len(data) is 16 / 5 = 3.2, the mean.

4

```
marks = [12, 150, 11, 14, 13]
print(sum(marks) / len(marks))
print(sorted(marks)[2])
```

Sorted, the marks are 11, 12, 13, 14, 150, so the middle one is 13.

5 **Data use, where a few people use a lot; House prices in a town with one mansion; Marks with one typed as 150**

When a few values are far from the rest, or one is a typo, the median tells the truth about the typical value.